RHO

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1 Target Audience and Problem Statement

Our Target Audience

- Hand Amputees
- Individuals who are unable to use their hands

Why?

These individuals are **severely impacted** when it comes to a hassle-free **passenger experience**, as a car has many elements. you need to use your hands for.

OUR GOAL

To design a highly **safe**, **seamless**, and **convenient** "hands-free passenger experience" for our target audience.

OUR PROBLEM STATEMENT

How can we provide a **hands-free passenger experience** in cars for individuals with **limited hand use?**

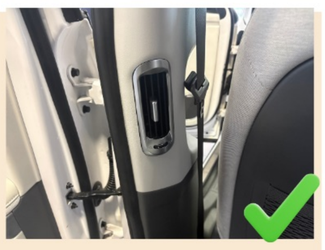
2 Empathy and Research

Putting ourselves in the shoes of our target audience in order to discover their struggles

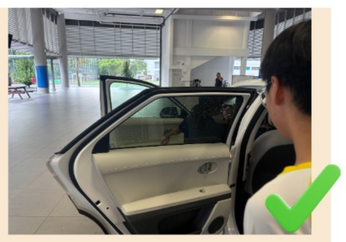
Difficulty of opening and closing doors

Difficulty on putting on seatbelts

We are not doing this due to safety concerns.



Difficulty in modifying the passenger air conditioning



Difficulty in window operation

3 Ideation

We wanted to design something **effective** and **genuinely useful** to our target audience.

"The single **most effective** way to make a service as reliable as possible is to make it as **simple as possible**"
~ Tom Limoncelli

Designing the **first-ever voice-automated system** for a car.

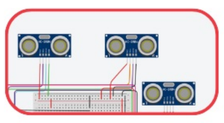
4 Our IMPORTANT Considerations

Ensuring a **SAFE and CONVENIENT** experience of our solution

Safety Issue	Problem Level (/10)	How did we tackle this?	Negated Risk Level (/10)
Receiving Command on Accident	5	Add a confirmation or a wake word (RHO)	2
Door hitting an object when opening/closing	9	Add a sensor to detect if the surroundings are closed. If they aren't, stop opening / closing the door	3
User does not know what is happening in the system	7	Make the system give quick audio responses back to the user	3
Door getting opened by someone else	8	The door will only open whenever the car key is near to the door	1

5 How RHO works

INPUT



KEY Detection
Ensures vehicle can only be used by their rightful owner

Enhances SAFETY

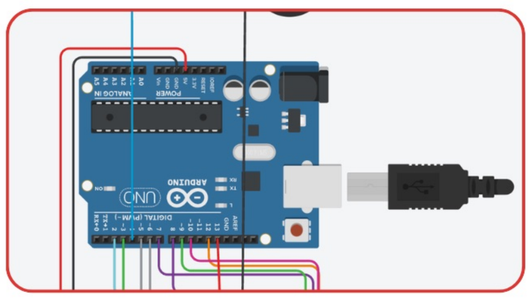
Voice Activation

- Waits for the wake word "RHO"
- Listens to the following command of user

Atmosphere Analysis
Safety is important! Ensures optimal conditions for the required function

Look at the design! All three ultrasonics are to ensure maximum safety.

PROCESS



Coding Softwares
PYTHON was used to code the voice activation

C++ was used to control the elements

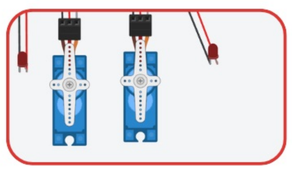
VALIDATION
Ensures that the command is a valid command recognised by the system!

SAFETY CHECKS
To ensure safety, operation will only start/continue if it is SAFE !
(e.g. no obstruction)

OUTPUT

Sends an Auditory Response Back
Ensures user has complete understanding of the whole process

Outputs Function
Controls the car based on the outpu through our electronic systems. Only if it passed all the safety and validation checks!



SIMPLE. YET EFFECTIVE.

BENEFITS & IMPACT

It can **benefit anybody** who can't use their hands, such as **carrying shopping bags.**

Easy to use due to its simple nature

Enhances passenger experience for our target audience in many different ways